

CALCULATION AND SIMPLIFICATION

eg $0.8666 \dots = 0.8\bar{6} = \frac{86-8}{90} = \frac{78}{90}$

$0.53131 \dots = 0.5\bar{31} = \frac{531-5}{990} = \frac{526}{990}$

$0.4377 \dots = 0.4\bar{37} = \frac{437-43}{900} = \frac{234}{900}$

$2.0\bar{4} = 2 + 0.\bar{4} = 2 + \frac{4}{9} = \frac{22}{9}$

Special Section:

37×3211

$11 \times 13 = 143$

$11 \times 17 = 187$

$11 \times 19 = 209$

$13 \times 17 = 221$

$16 \times 14 = 224$

$11 \times 12 = 132$

$24 \times 25 = 600$

$125 \times 8 = 1000$

$625 \times 8 = 5000$

Q. how to find sq.?

① last digit 5 आये तो:-

$25^2 = (2 \times 3) | 5^2 = 625$

$99.5^2 = 99 \times 100 | 5^2 = 9900.25$

BASE 100:-

$(103)^2 = (103+3) | 3^2 = 10609$

$(98)^2 = (98-2) | 2^2 = 9604$

$(87)^2 = (87+13) | 13^2 = 24169 = 2569$

BASE 50:- $\rightarrow 100 \left(\frac{1}{2} \right) \times 50$

$54^2 = \frac{1}{2} (54+4) | 4^2 = 2916$

$64^2 = \frac{1}{2} (64+14) | 14^2 = 3916 = 4096$

$(42)^2 = \frac{1}{2} (42+8) | 8^2 = 1764$

BASE 200:- $2 \times 0 \text{ of } 100$

$(217)^2 = 2(217+17) | 17^2 = 46089$

$(187)^2 = 2(187-13) | 13^2 = 34969$

if base 100 then multiple of ①

Base 50 $\rightarrow \frac{1}{2}$

Base 150 $\rightarrow \frac{3}{2}$

Base 200 $\rightarrow 2$

Base 250 $\rightarrow \frac{5}{2}$

Base 300 $\rightarrow 3$

Base 350 $\rightarrow \frac{7}{2}$

Base 450 $\rightarrow \frac{9}{2}$

Base $n^2 (50+n)^2 (100+n)^2 (200+n)^2$

CONDITION of a perfect sq.

① if a number ends with 2, 3, 7, 8 it can't be a perfect sq.

② if last digit is 9 then ten's digit is always 2.

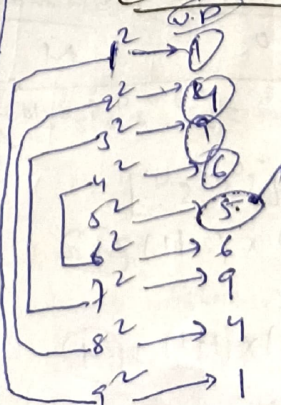
③ NO number can be a perfect sq. unless its digital sum is 0, 1, 4, 7 or 9.

④ last 2 digits of a perfect sq. NO 2 last two digits of the sq. of NUS BW (11024).

$N \begin{cases} 2K \text{ (even)} \rightarrow n^2 = (2K)^2 = 4K^2 \rightarrow 4(\text{even}) \\ 2K+1 \text{ (odd)} \rightarrow n^2 = (2K+1)^2 = 4K^2 + 4K + 1 = 4K(K+1) + 1 \end{cases}$

Remainder ①

SQUARE ROOTS



Step 1: $\sqrt{5329}$

① check whose last digit is 9 from 5329.

$3^2 \rightarrow 9$

② from step last two digit (ie 29)

③ which no. is perfect sq. near the leftward 53 ie $7^2 \rightarrow 49$

(iv) either 73 या 75 के निकटतम पूर्णांक में बदलकर जायेंगे:-

73 $\rightarrow 75^2 = 5625$ ($\because 5329 < 5625$)
 75
 then ans must be 75

ie $\rightarrow 73$ Ans

$\sqrt{\text{perfect sq.}}$ = Rational No.

CUBE ROOTS

- u.p
- $1^3 \rightarrow 1$
 - $2^3 \rightarrow 8$
 - $3^3 \rightarrow 27$
 - $4^3 \rightarrow 64$
 - $5^3 \rightarrow 125$
 - $6^3 \rightarrow 216$
 - $7^3 \rightarrow 343$
 - $8^3 \rightarrow 512$
 - $9^3 \rightarrow 729$

Steps:-

(i) from 493039 remove u.p 3

$7^3 \rightarrow$ u.p = 7

(ii) from step 1 (3) diff (039)

(iii) left nos. remaining is a perfect cube.

ie- 493

$7^3 = 343$ (near)

(iv)

$\sqrt[3]{493039} = 79$ Ans

DECIMAL BASED

$43600 = 436 \times 10^2$

$4.36 = \frac{436}{10^2} = 436 \times 10^{-2}$

$0.41937 = 41937 \times 10^{-5}$

$0.011 \times 10^3 = 0.011 \times 10^3$

$0.15 \times 0.008 = 0.0012$

(if (5×2) भाज्य हो एक दशमलव काटो जाता है).

(if $(5^2 \times 2^2)$ भाज्य हो दो दशमलव काटो जाता है).

$23^2 = 529$

$2.3^2 = 5.29$

$0.23^2 = 0.0529$

$0.00013^2 = 0.0000000169$

$\sqrt{881} = 29$

$\sqrt{3.61} = 1.9$

$\sqrt{0.000361} = 0.019$

BBP

$1 + 2 + 1 = 2^2$
 $1 + 2 + 3 + 2 + 1 = 3^2$
 $1 + 2 + 3 + 4 + 3 + 2 + 1 = 4^2$
 $\sqrt{1 + 2 + 1} = 2$
 $\sqrt{1 + 2 + 3 + 2 + 1} = 3$
 $\sqrt{1 + 2 + 3 + 4 + 3 + 2 + 1} = 4$

$1^2 = 1$

$11^2 = 121$

$111^2 = 12321$

$1111^2 = 1234321$

$11111^2 = 123454321$

ONLY for 6 8 9 :-

$3^2 = 09$

$33^2 = 1089$

$333^2 = 110889$

$33333^2 = 1111088889$

$9^2 = 81$

$99^2 = 9801$

$999999^2 = 999998000001$

BBP

$3^3 + 4^3 + 5^3 = 6^3$
 $1^3 + 6^3 + 8^3 = 9^3$

$3333^3 + 4444^3 + 5555^3 = 6666^3$

Algebra based simplification formulas

$a^2 - b^2 = (a+b)(a-b)$

$\frac{a^2 - b^2}{a+b} = a-b$

$\frac{a^2 - b^2}{a-b} = a+b$

$\sqrt{1234567654321} = 111111$

$6^2 = 36$

$66^2 = 4356$

$666^2 = 443556$

Formula's :-

$$\phi (a+b)^2 = a^2 + b^2 + 2ab$$

$$\phi \quad (a-b)^2 = a^2 + b^2 - 2ab.$$

$$\phi \quad (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

$$(a+b)^2 - (a-b)^2 = 4ab.$$

$$(a+b)^2 = (a-b)^2 + 4ab$$

$$\Phi \quad (a-b)^2 = (a+b)^2 - 4ab.$$

$$\begin{aligned} \text{+} \quad (a+b+c)^2 &= a^2+b^2+c^2+2ab+2bc+2ca \\ &= a^2+b^2+c^2+2[ab+bc+ca]. \\ \text{+} \quad (a+b-c)^2 &= a^2+b^2+c^2+2ab-2bc-2ca \\ &= a^2+b^2+c^2+2[ab-bc-ca] \end{aligned}$$

$$\begin{aligned} (a+b)^3 &= a^3 + b^3 + 3a^2b + 3ab^2 \\ &= a^3 + b^3 + 3ab(a+b) \end{aligned}$$

$$\begin{aligned} a^3 + b^3 &= (a+b)^3 - 3ab(a+b) \\ &= (a+b) \cdot [(a+b)^2 - 3ab] \\ &= (a+b) [a^2 + b^2 - ab] \text{ ans } \end{aligned}$$

$$(a-b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$$

$$= a^3 - b^3 - 3ab(a-b)$$

$$\begin{aligned} a^3 - b^3 &= (a-b)^3 + 3ab(a-b) \\ &= (a-b) \cdot [(a-b)^2 + 3ab] \\ &= (a-b) \cdot [a^2 + b^2 + ab]^{mp} \end{aligned}$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

FLAPPER FACTORY

$$\textcircled{2} \quad \frac{1}{\frac{1}{1+\frac{1}{1+\frac{1}{\frac{1}{5}}}}} = \frac{1}{1+\frac{1}{1+\frac{1}{\frac{1}{5}}}} = \frac{1}{1+\frac{1}{1+\frac{5}{1}}} = \frac{1}{1+\frac{1}{1+5}} = \frac{1}{1+\frac{1}{6}} = \frac{1}{\frac{7}{6}} = \frac{6}{7}$$

→ 2 5 7 12 19

$\textcircled{A}$

$N = D \quad (1P)$

$$\begin{array}{r} 1 \\ \hline 1 - 1 \\ 1 - 1 \\ 1 - 1 \\ 1 - 1 \\ 1 - 3 \\ 8 \end{array}$$

$2 = \frac{-8}{-5} = \textcircled{\frac{8}{5}}$

$\hookrightarrow 385 - 3$

④ $\frac{1}{1+1} = \frac{1}{2}$

Note—
if $1 + \frac{1}{1-1}$
 $\frac{3}{8}$
then reverse
the N&D.

27 5 12 7 19 26

$$\frac{1}{2+1} = \frac{68}{157}$$

(a)

$\begin{array}{r} 47 \\ 5 \end{array}$

$\begin{array}{r} 15 \\ 1 \end{array}$

$\begin{array}{r} 21 \\ 1 \end{array}$

$\begin{array}{r} 68 \\ N \end{array}$

$\begin{array}{r} 157 \\ D \end{array}$

$$\textcircled{a} \frac{1}{2+1} = \frac{131}{308}$$

$$\begin{array}{r} 54 \overline{) 11746} \\ \underline{108} \\ 94 \\ \underline{90} \\ 46 \\ \underline{45} \\ 16 \\ \underline{15} \\ 16 \end{array}$$

$$\textcircled{*} \frac{1}{2+3} \quad 2 \frac{32}{84} = \frac{11}{29}$$

$$\rightarrow \frac{1}{5} \times \frac{4}{7} \times \frac{3}{2} \times \frac{2}{3} \times \frac{1}{4} = \frac{1}{84}$$

38 →

$$\textcircled{*} (1+\frac{1}{2})(1+\frac{1}{3})(1+\frac{1}{4}) \dots (1+\frac{1}{80}) = \frac{81}{2}$$

$$\hookrightarrow \frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \dots \times \frac{81}{80} = \frac{81}{2}$$

$$\textcircled{*} (1-\frac{1}{3})(1-\frac{1}{6})(1-\frac{1}{7}) \dots (1-\frac{1}{75}) = \frac{4}{25}$$

$$\hookrightarrow \frac{2}{3} \times \frac{5}{6} \times \frac{6}{7} \times \dots \times \frac{74}{75} = \frac{4}{25}$$

$$\textcircled{*} (1-\frac{1}{35^2})(1-\frac{1}{36^2}) \dots (1-\frac{1}{209^2})$$

$$\left[(1+\frac{1}{35})(1+\frac{1}{36}) \dots (1+\frac{1}{209}) \right] \left[(1-\frac{1}{35})(1-\frac{1}{36}) \dots (1-\frac{1}{209}) \right]$$

$$\hookrightarrow \frac{210}{35} \times \frac{34}{209} = \frac{204}{209}$$

SERIES

$$\textcircled{*} \left[\frac{1}{3 \times 7} + \frac{1}{7 \times 11} + \frac{1}{11 \times 15} + \dots + \frac{1}{899 \times 903} \right]$$

$$= \frac{1}{4} \left[\frac{4}{3 \times 7} + \frac{4}{7 \times 11} + \frac{4}{11 \times 15} + \dots + \frac{4}{899 \times 903} \right]$$

$$= \frac{1}{4} \left[\frac{7-3}{3 \times 7} + \frac{11-7}{7 \times 11} + \frac{15-11}{11 \times 15} + \dots + \frac{903-899}{899 \times 903} \right]$$

$$= \frac{1}{4} \left[\frac{1}{3} - \frac{1}{7} + \frac{1}{7} - \frac{1}{11} + \frac{1}{11} - \frac{1}{15} + \dots + \frac{1}{899} - \frac{1}{903} \right]$$

$$= \frac{1}{4} \left[\frac{1}{3} - \frac{1}{903} \right] = \frac{25}{301}$$

$$38 \rightarrow \frac{1}{\dots} \left[\frac{1}{5} - \frac{1}{903} \right]$$

diff b/w two consecutive terms cancel

SODMAS

END